

1. INTRODUCTION

1.1 AIM OF THE PROJECT

The System is a comprehensive solution designed to ensure the safety of children by incorporating cutting-edge technology. The project integrates essential components, including temperature and pulse sensors, GPS, a panic button, and GSM communication.

1.2 OBJECTIVE OF THE PROJECT

The objective of the project is to ensure the safety and well-being of children by leveraging advanced technology. Through the integration of temperature and pulse sensors, GPS, a panic button, and GSM communication, the system aims to provide real-time monitoring of vital signs and location information. In the event of any abnormal sensor readings or activation of the panic button, the system will promptly send alerts via GSM, enabling swift response and intervention. This innovative project seeks to empower parents, guardians, and caregivers with a reliable and efficient tool for remote tracking and monitoring, contributing to the enhanced safety and security of children in various environments.

2. SYSTEM ANALYSIS

FEASIBILITY STUDY

A feasibility study for a child monitoring and tracking system would assess the practicality and viability of implementing such a system. Evaluate the availability of technology required for tracking and monitoring children, such as GPS tracking devices, communication modules, and sensor technology.

2.1 EXISTING SYSTEM

- In the existing methods for ensuring the safety of children often rely on manual supervision and periodic check-ins.
- Traditional approaches lack real-time data on vital signs and location, making it challenging for parents and caregivers to promptly respond to potential emergencies.
- Moreover, conventional tracking methods may not offer a comprehensive and integrated solution, leaving room for uncertainties in monitoring the well-being and whereabouts of children.
- The need for a more sophisticated system that combines advanced sensor technologies, GPS, and instant communication through GSM arises from the limitations of existing systems, aiming to bridge the gap and provide a proactive approach to child safety.

2.2 PROPOSED SYSTEM

The proposed Child Tracking and Monitoring System is a technologically advanced solution designed to overcome the limitations of existing methods. Integrating key components such as temperature and pulse sensors, GPS, a panic button, and GSM communication, the system ensures a comprehensive approach to child safety. When a sensor detects anomalies or if the panic button is activated, the system triggers an immediate message through GSM, enabling rapid response and intervention. Real-time monitoring of a child's vital signs, facilitated by temperature and pulse sensors, enhances health surveillance. The GPS component allows precise tracking of the child's location, providing accurate and timely information to parents and caregivers. This automated and proactive system not only addresses the shortcomings of traditional methods but also establishes a robust framework for ensuring the security and well-being of children in diverse settings.

3. SYSTEM REQUIREMENTS

3.1 HARDWARE REQUIREMENTS

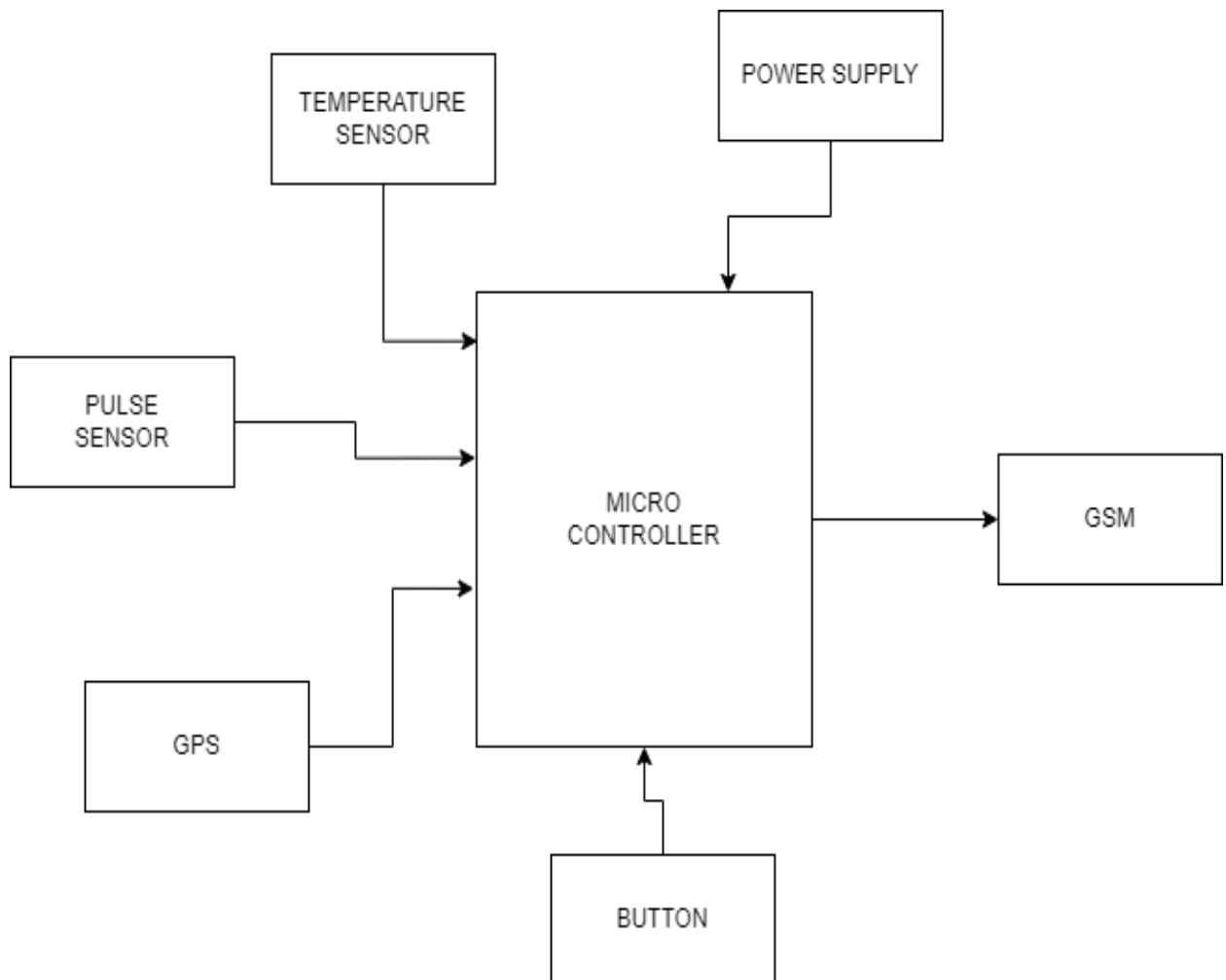
- Microcontroller
- Temperature sensor
- Pulse sensor
- GPS
- Button
- GSM

3.2 SOFTWARE REQUIREMENTS

- Arduino IDE
- Embedded C

4. SYSTEM DESIGN

4.1 SYSTEM ARCHITECTURE



5. MODULES DESCRIPTION

1. PULSE SENSOR
2. LM35 - TEMPERATURE SENSOR
3. GPS
4. PANIC BUTTON
5. GSM

1. PULSE SENSOR



- This The Heartbeat rate information knowing is very useful while doing exercise, studying, etc. But, the heartbeat rate can be complicated to calculate.
- To overcome this problem, the pulse sensor or heartbeat sensor is used.
- Heart Rate Monitoring: The pulse sensor detects the heartbeat of the child by measuring changes in blood volume in the capillaries under the skin. This data can be continuously collected and transmitted to a central monitoring unit.

- **Health Monitoring:** Monitoring a child's heart rate can provide valuable insights into their overall health and well-being. Sudden changes in heart rate could indicate potential health issues such as fever, dehydration, or stress.
- **Safety:** In situations where a child may be at risk of sudden health problems, such as seizures or cardiac issues, the pulse sensor can provide real-time alerts to caregivers or medical professionals, allowing for prompt intervention.
- **Activity Tracking:** Heart rate data can also be used to track the child's activity levels and exertion during physical activities. This information can help parents ensure their child is getting enough exercise or detect if they are overexerting themselves.
- **Sleep Monitoring:** Monitoring heart rate during sleep can provide insights into the quality of the child's sleep and detect any irregularities that may indicate sleep disorders or disturbances.

2. LM35 - TEMPERATURE SENSOR



- **Monitoring Environment Temperature:** The LM35 is a precision integrated circuit temperature sensor that provides an analog output voltage proportional to the surrounding temperature. This feature can be utilized to monitor the ambient temperature around the child, ensuring they are not exposed to extreme temperatures that could pose health risks.

- **Detecting Fever or Hyperthermia:** If the system is designed to monitor the child's health parameters, the LM35 can be used to detect elevated body temperatures, indicating fever or hyperthermia. This information can trigger alerts to caregivers or healthcare providers, enabling prompt intervention.
- **Preventing Overheating or Hypothermia:** Monitoring the child's body temperature can also help prevent overheating or hypothermia, especially in environments where temperature regulation is crucial, such as in a car seat or during outdoor activities.
- **Enhancing Safety Features:** Integrating temperature sensing capabilities can enhance the safety features of the tracking system. For instance, if the temperature inside a vehicle becomes dangerously high (e.g., due to being left in a parked car), the system can alert caregivers to take immediate action to prevent heat-related illnesses.
- **Customized Alerts and Notifications:** By integrating temperature monitoring with the tracking system, caregivers can receive customized alerts and notifications based on predefined temperature thresholds. This allows for proactive measures to be taken in response to temperature fluctuations.

3. GPS



- **Real-time Location Tracking:** GPS enables parents or guardians to track the precise location of their child in real-time using GPS-enabled devices such as smartphones, wearable trackers, or dedicated GPS trackers. This helps ensure

the child's safety and provides peace of mind to caregivers, especially in situations where the child might be out of sight or in unfamiliar surroundings.

- Geo-fencing: GPS technology allows for the creation of virtual boundaries known as geo-fences. Parents can set up specific geographical areas as safe zones, such as home, school, or a park. When the child enters or exits these predefined areas, the system can automatically send alerts to the parent's device, notifying them of the child's movements.
- Emergency Response: In case of an emergency or if the child gets lost, GPS tracking enables quick and accurate location identification, facilitating a prompt response from caregivers or emergency services. This can be crucial in situations where immediate intervention is required to ensure the safety and well-being of the child.
- Historical Location Data: GPS tracking systems often store historical location data, allowing parents to review the child's travel routes and locations visited over a specified period. This feature can be useful for tracking daily routines, identifying patterns, and ensuring that the child is sticking to planned activities or schedules.
- Navigation Assistance: Some advanced child tracking systems may offer navigation features to guide parents or guardians to the child's current location, especially if they are in an unfamiliar area. This can be particularly helpful in situations where the child needs assistance or if the caregiver needs to quickly locate them.

4. PANIC BUTTON



In a child monitoring and tracking system, a panic button serves as a critical safety feature. Its primary function is to allow the child to quickly and discreetly send an alert in case of an emergency or when they feel threatened or unsafe. When activated, the panic button sends an immediate notification to a predefined contact or monitoring center, informing them of the child's distress and providing their location coordinates if GPS tracking is integrated into the system.

The use of a panic button provides peace of mind to both parents and guardians, as well as the child themselves, knowing that help can be summoned swiftly in case of any danger or emergency situation. It can be particularly beneficial in scenarios such as abduction attempts, accidents, medical emergencies, or when the child finds themselves in an uncomfortable or threatening situation.

5. GSM



- **Location Tracking:** GSM technology enables the tracking device worn by the child to communicate its location to a central server or to a parent's device. This can be achieved through triangulation of signals from nearby cell towers, allowing for relatively accurate determination of the child's location.
- **Communication:** Many child monitoring systems incorporate GSM modules that allow for two-way communication between the child and their parent or caregiver. This can be crucial in emergencies or simply for keeping in touch. Parents can call or send messages to the device worn by the child, and vice versa.
- **Alerts and Notifications:** GSM connectivity enables the system to send alerts and notifications to parents or caregivers in real-time. For example, if the child exits a predefined safe zone, such as their school or home, the system can automatically send an alert to the parent's phone.
- **Remote Management:** GSM connectivity allows parents or caregivers to remotely manage the tracking device worn by the child. This may include adjusting settings, updating software, or even remotely activating features such as a panic button or microphone for listening in on the child's surroundings.
- **Backup Communication:** In case other forms of communication (such as Wi-Fi or Bluetooth) are unavailable, GSM provides a reliable backup option for maintaining connectivity between the tracking device and the parent's device or the central monitoring system.

6. HARDWARE DESCRIPTION

1. PULSE SENSOR

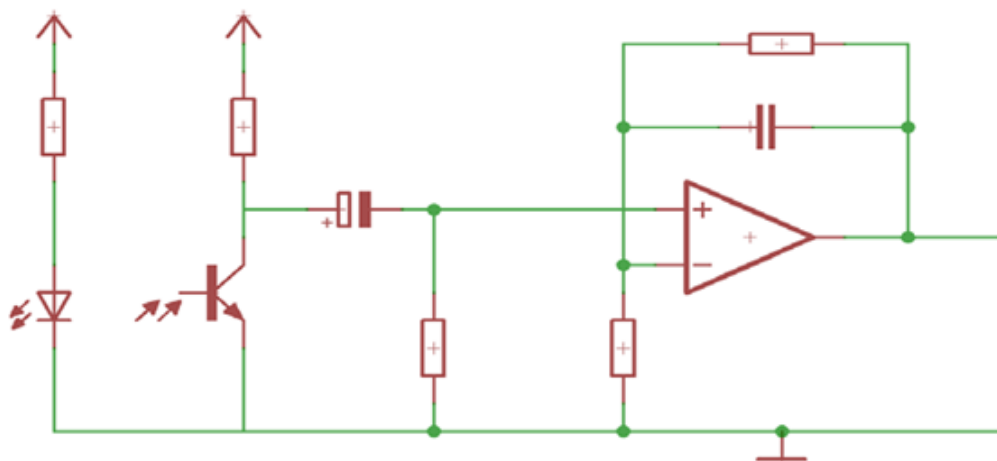
The basic heartbeat sensor consists of a light emitting diode and a detector like a light detecting resistor or a photodiode. The heart beat pulses causes a variation in the flow of blood to different regions of the body. When a tissue is illuminated with the light source, i.e. light emitted by the led, it either reflects (a finger tissue) or transmits the light (earlobe). Some of the light is absorbed by the blood and the transmitted or the reflected light is received by the light detector. The amount of light absorbed depends on the blood volume in that tissue. The detector output is in form of electrical signal and is proportional to the heart beat rate.

This signal is actually a DC signal relating to the tissues and the blood volume and the AC component synchronous with the heart beat and caused by pulsatile changes in arterial blood volume is superimposed on the DC signal. Thus the major requirement is to isolate that AC component as it is of prime importance.

To achieve the task of getting the AC signal, the output from the detector is first filtered using a 2 stage HP-LP circuit and is then converted to digital pulses using a comparator circuit or using simple ADC. The digital pulses are given to a microcontroller for calculating the heart beat rate, given by the formula-

$$\text{BPM(Beats per minute)} = 60 * f$$

Where f is the pulse frequency



Features

- Biometric Pulse Rate or Heart Rate detecting sensor
- Plug and Play type sensor
- Operating Voltage: +5V or +3.3V
- Current Consumption: 4mA
- Inbuilt Amplification and Noise cancellation circuit.
- Diameter: 0.625"
- Thickness: 0.125" Thick

Applications

- Sleep Tracking
- Anxiety monitoring
- Remote patient monitoring/alarm system
- Health bands
- Advanced gaming consoles

2. LM35 - TEMPERATURE SENSOR

The LM35 is an integrated circuit sensor that can be used to measure temperature with an electrical output proportional to the temperature (in °C)

The LM35 - An Integrated Circuit Temperature Sensor

Why Use LM35s To Measure Temperature?

- You can measure temperature more accurately than using a thermistor.
- The sensor circuitry is sealed and not subject to oxidation, etc.
- The LM35 generates a higher output voltage than thermocouples and may not require that the output voltage be amplified.

What Does an LM35 Do? How does it work?

- It has an output voltage that is proportional to the Celsius temperature.
- The scale factor is .01V/oC
- The LM35 does not require any external calibration or trimming and maintains an accuracy of +/-0.4 oC at room temperature and +/- 0.8 oC over a range of 0 oC to +100 oC.
- Another important characteristic of the LM35DZ is that it draws only 60 micro amps from its supply and possesses a low self-heating capability. The sensor self-heating causes less than 0.1 oC temperature rise in still air.

The LM35 comes in many different packages, including the following.

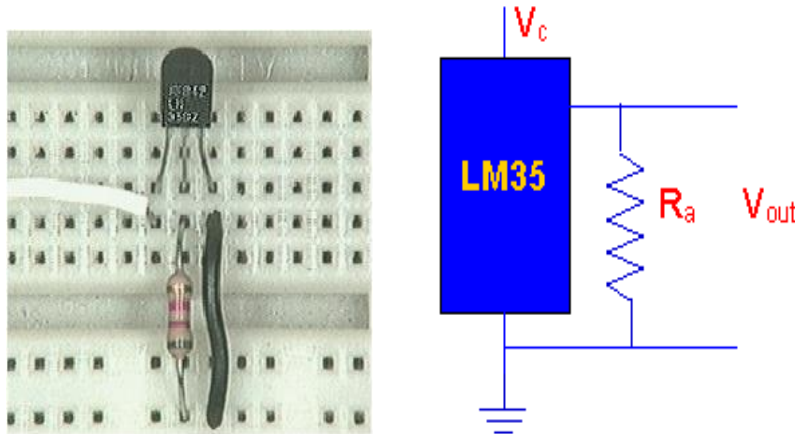
- TO-92 plastic transistor-like package,
- TO-46 metal can transistor-like package
- 8-lead surface mount SO-8 small outline package
- TO-202 package. (Shown in the picture above)

How Do You Use An LM35? (Electrical Connections)

- Here is a commonly used circuit. For connections refer to the picture above.
- In this circuit, parameter values commonly used are:
 - $V_c = 4 \text{ to } 30\text{v}$
 - 5v or 12 v are typical values used.
 - $R_a = V_c / 10^{-6}$
 - Actually, it can range from 80 KW to 600 KW , but most just use 80 KW.

Here is a photo of the LM 35 wired on a circuit board.

- The white wire in the photo goes to the power supply.
- Both the resistor and the black wire go to ground.
- The output voltage is measured from the middle pin to ground.



What Can You Expect When You Use An LM35?

- You will need to use a voltmeter to sense V_{out} .
- The output voltage is converted to temperature by a simple conversion factor.
- The sensor has a sensitivity of $10\text{mV} / ^\circ\text{C}$.
- Use a conversion factor that is the reciprocal, that is $100\text{V} / ^\circ\text{C}$.
- The general equation used to convert output voltage to temperature is:
 - $\text{Temperature } (^\circ\text{C}) = V_{out} * (100 ^\circ\text{C/V})$
 - So if V_{out} is 1V , then, $\text{Temperature} = 100 ^\circ\text{C}$
 - The output voltage varies linearly with temperature.

3. GPS

GPS satellites circle the Earth twice a day in a precise orbit. Each satellite transmits a unique signal and orbital parameters that allow GPS devices to decode and compute the precise location of the satellite. GPS receivers use this information and trilateration to calculate a user's exact location. Essentially, the GPS receiver measures the distance to each satellite by the amount of time it takes to receive a transmitted signal. With distance measurements from a few more satellites, the receiver can determine a user's position and display it. To calculate your 2-D position (latitude and longitude) and track movement, a GPS receiver must be locked on to the signal of at least 3 satellites. With 4 or more satellites in view, the receiver can determine your 3-D position (latitude, longitude and altitude). Generally, a GPS receiver will track 8 or more satellites, but that depends on the time of day and where

you are on the earth. Once your position has been determined, the GPS unit can calculate other information, such as:

- Speed
- Bearing
- Track
- Trip dist
- Distance to destination

What's the signal?

GPS satellites transmit at least 2 low-power radio signals. The signals travel by line of sight, meaning they will pass through clouds, glass and plastic but will not go through most solid objects, such as buildings and mountains. However, modern receivers are more sensitive and can usually track through houses.

A GPS signal contains 3 different types of information:

- Pseudorandom code is an I.D. code that identifies which satellite is transmitting information. You can see which satellites you are getting signals from on your device's satellite page.
- Ephemeris data is needed to determine a satellite's position and gives important information about the health of a satellite, current date and time.
- Almanac data tells the GPS receiver where each GPS satellite should be at any time throughout the day and shows the orbital information for that satellite and every other satellite in the system.

1. NEO-6M GPS Chip

The heart of the module is a NEO-6M GPS chip from u-blox. It can track up to 22 satellites on 50 channels and achieves the industry's highest level of sensitivity i.e. -161 dB tracking, while consuming only 45mA supply current. The u-blox 6 positioning engine also boasts a Time-To-First-Fix (TTFF) of under 1 second. One of the best features the chip provides is Power Save Mode (PSM). It allows a reduction in system power consumption

by selectively switching parts of the receiver ON and OFF. This dramatically reduces power consumption of the module to just 11mA making it suitable for power sensitive applications like GPS wristwatch. The necessary data pins of NEO-6M GPS chip are broken out to a "0.1" pitch headers. This includes pins required for communication with a microcontroller over UART.

Note: - The module supports baud rate from 4800bps to 230400bps with default baud of 9600.



2. Position Fix LED Indicator

There is an LED on the NEO-6M GPS Module which indicates the status of Position Fix. It'll blink at various rates depending on what state it's in

No Blinking ==> means It is searching for satellites

Blink every 1s – means Position Fix is found



3. 3.3V LDO Regulator

The operating voltage of the NEO-6M chip is from 2.7 to 3.6V. But, the module comes with MIC5205 ultra-low dropout 3V3 regulator from MICREL. The logic pins are also 5-volt tolerant, so we can easily connect it to an Arduino or any 5V logic microcontroller without using any logic level converter.



4. Battery & EEPROM

- The module is equipped with an HK24C32 two wire serial EEPROM. It is 4KB in size and connected to the NEO-6M chip via I2C. The module also contains a rechargeable button battery which acts as a super-capacitor.
- An EEPROM together with battery helps retain the battery backed RAM (BBR). The BBR contains clock data, latest position data (GNSS or bit data) and module configuration. But it is not meant for permanent data storage.
- As the battery retains clock and last position, time to first fix (TTFF) significantly reduces to 1s. This allows much faster position locks.
- Without the battery the GPS always cold-start so the initial GPS lock takes more time. The battery is automatically charged when power is applied and maintains data for up to two weeks without power.

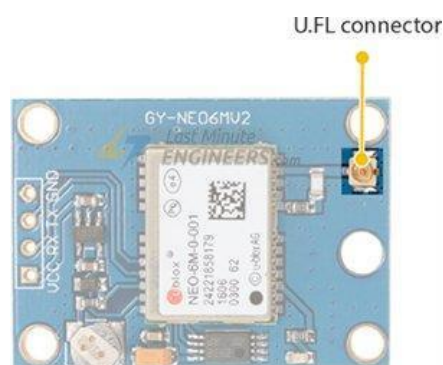


5. Antenna

An antenna is required to use the module for any kind of communication. So, the module comes with a patch antenna having -161 dBm sensitivity.



You can snap-fit this antenna to small U.FL connector located on the module.



Patch antenna is great for most projects. But if you want to achieve more sensitivity or put your module inside a metal case, you can also snap on any 3V active GPS antenna via the U.FL connector.

6. Pinout



- GND is the Ground Pin and needs to be connected to GND pin on the Arduino.
- TxD (Transmitter) pin is used for serial communication.
- RxD (Receiver) pin is used for serial communication.
- VCC supplies power for the module. You can directly connect it to the 5V pin on the Arduino.

4. PANIC BUTTON

Push Buttons are normally-open tactile switches. Push buttons allow us to power the circuit or make any particular connection only when we press the button. Simply, it makes the circuit connected when pressed and breaks when released. A push button is also used for triggering of the SCR by gate terminal. These are the most common buttons which we see in our daily life electronic equipment's.

Push Button Features

- Prevent flux rise by the insert-molded terminal
- Snap-in mount terminal
- Contact Bounce: MAX 5mS
- Crisp clicking by tactile feedback
- Dielectric Withstanding Voltage 250V AC for 1 minute

Technical Specifications

- Mode of Operation: Tactile feedback
- Power Rating: MAX 50mA 24V DC
- Insulation Resistance: 100Mohm at 100v
- Operating Force: 2.55 ± 0.69 N
- Contact Resistance: MAX 100mOhm
- Operating Temperature Range: -20 to +70 °C
- Storage Temperature Range: -20 to +70 °C

Applications

- Calculators
- Push-button telephones
- Kitchen appliances
- Magnetic locks
- Various other mechanical and electronic devices, home and commercials.

5. GSM

The SIM900A is a GSM/GPRS module designed for mobile communication applications. Operating on the GSM network, it supports both voice and data communication, enabling devices to make calls, send SMS messages, and connect to the internet via GPRS. With a SIM card for authentication, the module communicates through a serial interface, making it compatible with various microcontrollers and embedded systems. Operating within the 3.4V to 4.5V voltage range, the SIM900A is known for its low power consumption, making it suitable for battery-powered applications. The module is controlled using AT commands, allowing users to perform tasks like making calls or configuring network settings. Widely used in remote monitoring, tracking systems, and security applications, the SIM900A facilitates wireless connectivity for diverse electronic projects.

Applications

- Cellular Communication
- Robotics
- Mobile Phone Accessories
- Servers
- Computer Peripherals
- Automobile
- USB Dongles

7. SOFTWARE DESCRIPTION



- **Programming Microcontrollers:** Arduino IDE is primarily used to program Arduino microcontroller boards. These microcontrollers can be utilized in child monitoring and tracking systems to collect data from sensors, process information, and control actuators or devices.
- **Interface with Sensors and Modules:** Child monitoring systems often incorporate various sensors such as GPS modules, accelerometers, heart rate sensors, etc. Arduino IDE allows developers to write code to interface these sensors with the microcontroller board, enabling data collection and analysis.
- **Data Processing and Decision Making:** The IDE allows developers to write code that processes data collected from sensors. For instance, it could analyze GPS data to determine the child's location, or analyze accelerometer data to detect if the child has fallen.
- **Communication Protocols:** Arduino IDE supports various communication protocols such as Bluetooth, Wi-Fi, GSM, etc. These protocols can be utilized to transmit data from the child monitoring device to a parent's smartphone or a central monitoring station.
- **Control Mechanisms:** The Arduino code can incorporate control mechanisms based on the received data. For example, it could trigger an alarm if the child moves out of a predefined safe zone or send an alert if certain sensor readings indicate a potential danger.
- **Customization and Expansion:** Arduino IDE allows for easy customization and expansion of the monitoring system. Developers can add new features,

integrate additional sensors, or modify existing functionalities to meet specific requirements.

The Embedded C Programming Language:

- Embedded C is a variation of the C programming language tailored for programming embedded systems, which are specialized computing systems that are part of larger devices or systems.
- These systems typically have limited resources such as memory, processing power, and sometimes power consumption requirements.
- Embedded C is designed to be efficient, reliable, and suitable for the unique constraints and demands of embedded systems development.

That can be characterized by all of the following:

1. Close-to-Hardware Programming
2. Resource Management
3. Real-Time Constraints
4. Portability
5. Compiler Specifics
6. Low-Level Programming
7. Interrupt Handling
8. Debugging and Testing
9. Safety and Reliability
10. Continuous Learning

COMPILATION & EXECUTION

- **Start:** Execution begins when the Arduino board is powered on or reset.
- **Setup() Function:** The setup() function is called once at the beginning of the program and is used for initialization tasks.
- **Setup Hardware:** Within the setup() function, hardware components such as pins and peripherals are initialized.
- **Initialization of Variables and Libraries:** Variables are initialized, and any required libraries are included and initialized.
- **Loop() Function:** The loop() function is the main execution loop of the program and is continuously executed after the setup() function completes.
- **Read Inputs and Sensors:** Inputs from sensors or external devices are read within the loop() function.
- **Process Data and Perform Computations:** Data from inputs is processed, and any required computations are performed.
- **Control Outputs:** Based on the processed data, outputs such as LEDs, motors, or other actuators are controlled.
- **Delay (if used):** Optionally, a delay may be included to introduce a pause in the execution loop.
- **Repeat Loop:** The loop() function continues to execute repeatedly, cycling through the input processing, computation, output control, and delay steps.
- **End:** The program continues to run until the Arduino board is powered off or reset.

Uses OF Embedded C:

- Embedded C is extensively used in consumer electronics such as smartphones, digital cameras, portable media players, smart TVs, and home appliances.
- In industrial automation, Embedded C is used for controlling machinery, process automation, monitoring sensors, and interfacing with various industrial protocols.
- Embedded C is crucial in medical devices such as patient monitoring systems, infusion pumps, diagnostic equipment, and implantable devices.

- Embedded C is used in aerospace and defense applications for avionics systems, unmanned aerial vehicles (UAVs), radar systems, missile guidance systems, and satellite communication systems. These applications often require high-performance computing in resource-constrained environments.
- Embedded C is utilized in communication systems including routers, switches, network appliances, and modems for handling network protocols, data transmission, and managing network interfaces efficiently.
- Embedded C plays a significant role in IoT devices such as smart home devices, wearable technology, industrial sensors, and environmental monitoring systems. These devices often operate on low-power microcontrollers and require efficient code for battery-powered operation.

The Advantages of Embedded C

- **Efficiency:** Embedded C allows for efficient use of system resources such as memory and processing power. It is a low-level language, allowing programmers to directly control hardware resources, leading to optimized code execution.
- **Direct Hardware Access:** Embedded C provides direct access to hardware peripherals, such as GPIO pins, timers, and ADCs, allowing developers to precisely control hardware components. This level of control is crucial for interfacing with sensors, actuators, and other external devices.
- **Real-time Capabilities:** Many embedded systems require real-time responsiveness, where tasks must be completed within specific time constraints. Embedded C enables developers to write code with deterministic timing behavior, ensuring timely responses to external events.
- **Portability:** Embedded C code can be relatively easily ported across different microcontroller architectures and development environments. This flexibility allows developers to reuse code and migrate projects to new hardware platforms with minimal modifications.

- Low-Level Programming: Embedded C allows developers to work at a low level of abstraction, closer to the hardware, which is essential for tasks like device driver development, bootloader programming, and optimizing performance-critical code sections.

8. SAMPLE CODINGS

```
#include <SoftwareSerial.h>

#include <TinyGPS.h>

float lat =17.433426,lon = 78.447054; // create variable for latitude and longitude
object

SoftwareSerial gpsSerial(6,7);//rx,tx

TinyGPS gps; // create gps object

#include <SoftwareSerial.h>

SoftwareSerial mySerial(4,5);

int SW=2;

int pulse=A1;

int temp=A0;

int x,y,hb,z;

void setup(){

Serial.begin(9600); // connect serial

Serial.println("The GPS Received Signal:");

gpsSerial.begin(9600); // connect gps sensor

mySerial.begin(9600);

pinMode(SW,INPUT_PULLUP);

pinMode(temp,INPUT);

pinMode(pulse,INPUT);

}
```

```

void loop()

{

//while(gpsSerial.available()){ // check for gps data

// if(gps.encode(gpsSerial.read()))// encode gps data

// {

// gps.f_get_position(&lat,&lon); // get latitude and longitude

// Serial.print("Position: ");

// Serial.print("Latitude:");

// Serial.print(lat,6);

// Serial.print(";");

// Serial.print("Longitude:");

// Serial.println(lon,6); ;

// Serial.print(lat);

// Serial.print(" ");

// }

//}

```

```

int x=analogRead(temp);

int y=analogRead(pulse);

int hb=((y/10)+20);

int z=digitalRead(SW);

Serial.print("Panic button:");

Serial.println(z);

```

```

Serial.print("pulse:");

Serial.println(hb);

Serial.print("temperature:");

Serial.println(x);

delay(1000);

if (z== 0)
{
    mySerial.println("AT+CMGF=1"); //Sets the GSM Module in Text Mode

    delay(1000); // Delay of 1 second

    mySerial.println("AT+CMGS=\""+919514546514"\\r"); // Replace x with mobile
number

    delay(1000);

    mySerial.println("PANIC");// The SMS text you want to send

    delay(100);

    mySerial.println((char)26);// ASCII code of CTRL+Z for saying the end of sms to
the module

    delay(1000);

    while(gpsSerial.available()){ // check for gps data
    if(gps.encode(gpsSerial.read()))// encode gps data
    {

    gps.f_get_position(&lat,&lon); // get latitude and longitude

    Serial.print("Position: ");

    Serial.print("Latitude:");

```

```

Serial.print(lat,6);

Serial.print(";");

Serial.print("Longitude:");

Serial.println(lon,6); ;

Serial.print(lat);

Serial.print(" ");

}

}

Serial.println("women is danger please help me");


}


if (x>35)
{

mySerial.println("AT+CMGF=1"); //Sets the GSM Module in Text Mode

delay(1000); // Delay of 1 second

mySerial.println("AT+CMGS=\""+919514546514+"\r"); // Replace x with mobile
number

delay(1000);

mySerial.println("HIGH TEMPERATURE");// The SMS text you want to send

delay(100);

mySerial.println((char)26);// ASCII code of CTRL+Z for saying the end of sms to
the module

```

```

    delay(1000);

    while(gpsSerial.available()){ // check for gps data
if(gps.encode(gpsSerial.read()))// encode gps data
{
    gps.f_get_position(&lat,&lon); // get latitude and longitude

    Serial.print("Position: ");

    Serial.print("Latitude:");

    Serial.print(lat,6);

    Serial.print(";");

    Serial.print("Longitude:");

    Serial.println(lon,6); ;

    Serial.print(lat);

    Serial.print(" ");

}

}

    Serial.println("temp ");

}

if (hb>85)

{

    mySerial.println("AT+CMGF=1"); //Sets the GSM Module in Text Mode

    delay(1000); // Delay of 1 second

```



```

    mySerial.println("AT+CMGS=\""+919514546514+"\r"); // Replace x with mobile
number

    delay(1000);

    mySerial.println("PULSE,18.98308,10.6836");// The SMS text you want to send

    delay(100);

    mySerial.println((char)26);// ASCII code of CTRL+Z for saying the end of sms to
the module

    delay(1000);

    Serial.println("pulse value ");

    while(gpsSerial.available()){ // check for gps data
if(gps.encode(gpsSerial.read()))// encode gps data
{

gps.f_get_position(&lat,&lon); // get latitude and longitude

Serial.print("Position: ");

Serial.print("Latitude:");

Serial.print(lat,6);

Serial.print(";");

Serial.print("Longitude:");

Serial.println(lon,6); ;

Serial.print(lat);

Serial.print(" ");

}

}

```

```
}  
  
else  
  
{  
  
    Serial.println("normal");  
  
    delay(1000);  
  
    //SendMessage("PLSE HEALP ME");  
  
}  
  
}
```

9. SYSTEM TESTING AND MAINTENANCE

9.1. SYSTEM TESTING

Unit Testing:

- Test individual components/modules of the system to ensure they function as expected.
- Verify the correctness of algorithms, data processing, and input/output.

Integration Testing:

- Test the integration of individual components to ensure they work together seamlessly.
- Verify data flow and communication between modules.

System Testing:

- Test the system as a whole to ensure it meets the specified requirements.
- Verify user interfaces, system responses to various inputs, and overall functionality.
- Perform usability testing to ensure ease of use for parents and caregivers.
- Test different scenarios including normal usage, edge cases, and error conditions.

Performance Testing:

- Assess the system's performance under different loads and usage conditions.
- Measure response times, throughput, and resource usage.
- Identify and address any bottlenecks or performance issues.

Security Testing:

- Conduct security assessments to identify and address potential vulnerabilities.
- Test for authentication, authorization, data encryption, and secure communication.

- Ensure compliance with relevant privacy regulations (e.g., GDPR, COPPA).

User Acceptance Testing (UAT):

- Involve actual users (parents, caregivers) to validate the system meets their needs and expectations.
- Gather feedback and make necessary adjustments based on user input.

9.2 SYSTEM IMPLEMENTATION

Bug Fixes:

- Address any bugs or issues identified during testing or reported by users promptly.
- Prioritize and categorize bugs based on severity and impact on system functionality.

Updates and Upgrades:

- Regularly update the system to incorporate new features, improvements, and security patches.
- Ensure compatibility with new operating systems, devices, or platforms.
- Communicate updates to users and provide guidance on how to apply them.

Database Maintenance:

- Regularly back up and optimize the database to prevent data loss and ensure efficient performance.
- Implement data retention policies to manage storage space effectively.

Security Maintenance:

- Stay vigilant against emerging security threats and update security measures accordingly.
- Conduct regular security audits and penetration testing to identify and mitigate risks.

User Support:

- Provide ongoing support to users through various channels (e.g., helpdesk, documentation, FAQs).
- Address user inquiries, issues, and concerns promptly to maintain user satisfaction.

Monitoring and Analytics:

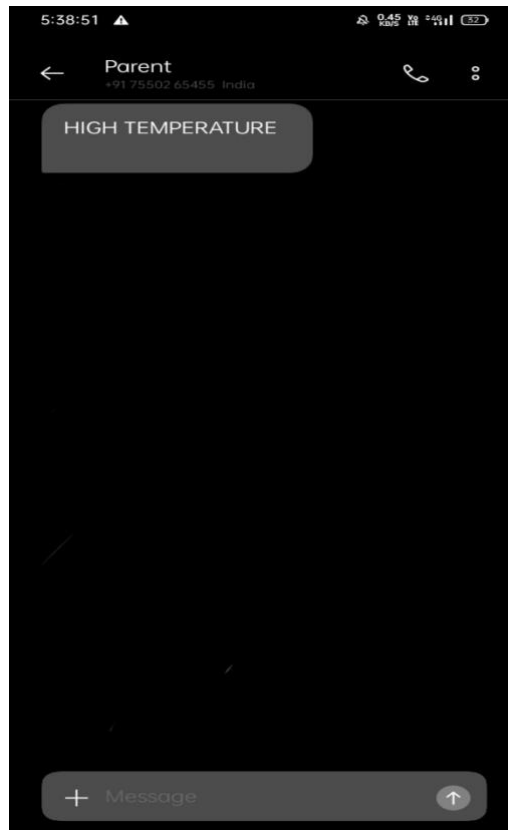
- Implement monitoring tools to track system performance, usage patterns, and potential issues.
- Use analytics to gain insights into user behavior and preferences, informing future improvements.

Compliance:

- Stay updated on relevant regulations and standards pertaining to child monitoring and data privacy.
- Ensure ongoing compliance with legal requirements through regular audits and adjustments.

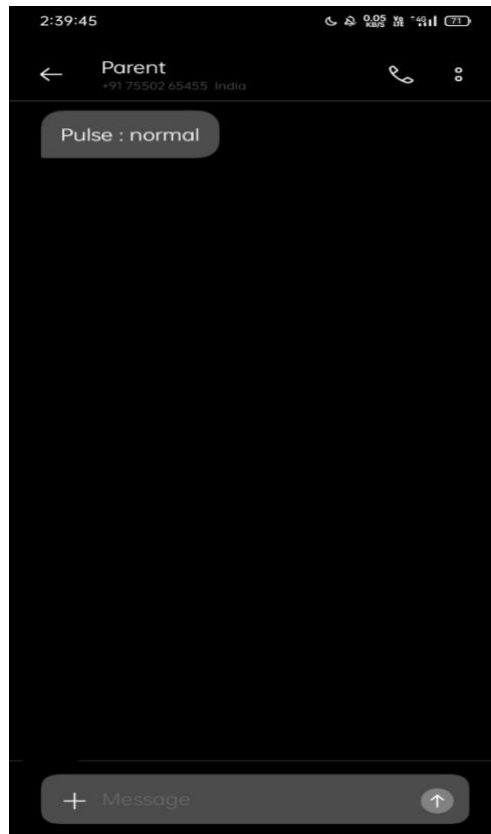
STEP 2:

Temperature sensor alert message send to parent through GSM.



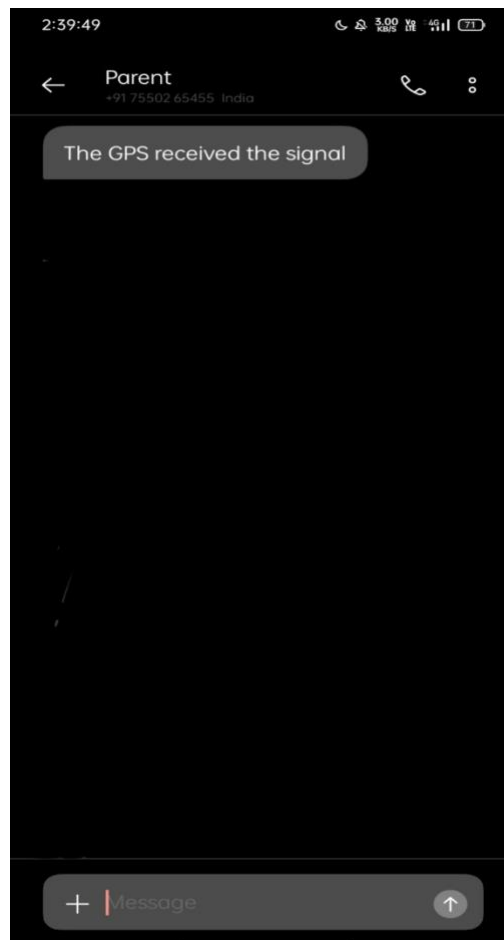
STEP 3:

Pulse sensor alert messages send to parent through GSM.



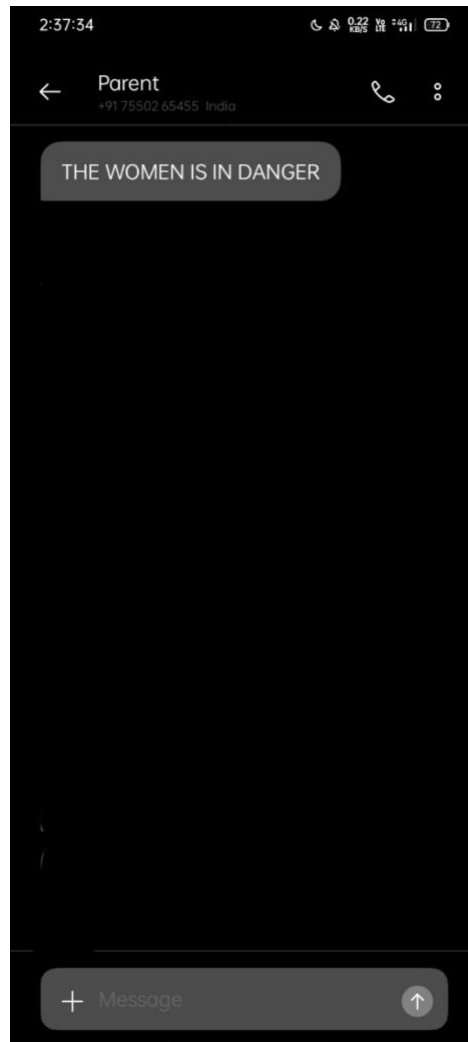
STEP 4:

GPS tracking signal is received to the parents through the GSM.



STEP 5:

Panic button alert message send to parent through GSM.



11. COST ESTIMATION

MATERIAL	COST	TOTAL
BROWSING	500	3000
TRAINING	500	3000
REPORT PREPARATION	250	2000
	TOTAL	8000

12. CONCLUSION

In conclusion, the project demonstrates a significant leap in child safety technology by seamlessly integrating temperature and pulse sensors, GPS, panic button, and GSM communication through a centralized microcontroller. This comprehensive approach ensures real-time monitoring of a child's health and precise tracking of their location. The system's ability to promptly send alerts through GSM in response to sensor detection enhances the speed and efficiency of emergency responses. By leveraging advanced components and technologies, this project provides parents and caregivers with a reliable, user-friendly solution for easily monitoring and tracking children. The Child Tracking and Monitoring System not only addresses the limitations of existing methods but sets a new standard in ensuring the safety and well-being of children in various environments.

13. BIBLIOGRAPHY

12.1 References

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14. USER MANUAL

Welcome to CHILD MONITORING AND TRACKING SYSTEM USING IOT, here you can know full specification about the system.

These user manual guide you to use the application.

- Getting Started
- GSM connectivity
- Adding Your Child's Device
- Using the Dashboard
- Tracking Your Child
- Sense the pulse
- Getting Alerts
- Sense the temperature
- Panic button
- Legal Information